Deep Learning

Session 1: Deep learning in public policy

September 3, 2024

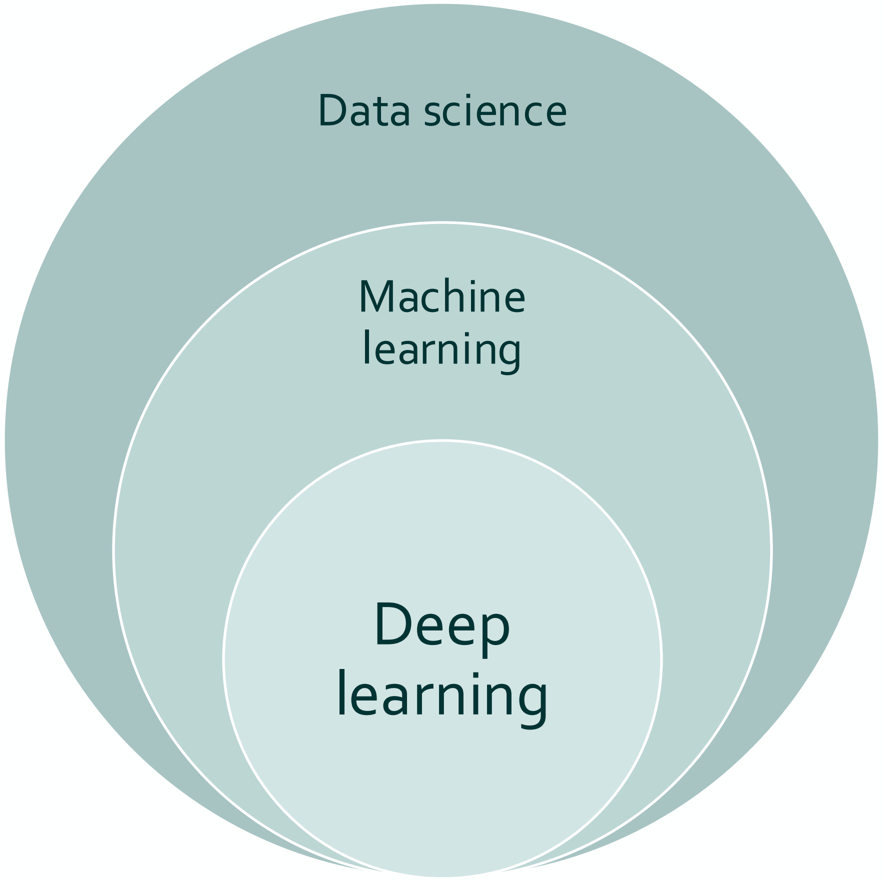
This session will be recorded for in-class use.

What this course is about

Hertie School

Deep learning

- Neural networks with more hidden layers
- Achieve top performance on many complex tasks including computer vision, speech recognition, games, etc.
- Theory around since 1950s and 1960s
- 1980s: backpropagation algorithm, classifying handwritten digits, medical image segmentations
- "AI winter": computationally intractable for real-world problems
- Around 2000 advances in computing and deep learning methods
- Really took off around 2010 with GPU's used for training
- Requires large datasets and large computational resources
- Deep learning models and librarys update and change very fast



Deep learning innovations

Newest developments

Deep learning innovations

Generative

Hertie School

2022: Generating text - GPT-3

Playground

Load a preset... | Save | View code | Share | ...

As a student in the Masters of Data Science and Public Policy in Berlin, I was curious on how the Corona Virus affected the dining habits of Germans.

I used the data of "Statista" to analyze the data of "The number of times people in Germany ordered food delivery or take out in 2020, by type of food (in 1,000s)."

I found that there was a significant increase in the number of people ordering food delivery or take out in 2020, especially for pizzas and Chinese food.

Mode

Model

text-davinci-002

Temperature 0.7

Maximum length 256

Stop sequences

Enter sequence and press Tab

- GPT-3 is a large language model developed by OpenAI
- 175 billion parameters trained on ~500 billion words
- Capable of write prose, academic papers, math, coding in several languages
- The source code is not open
- This slide: 2022 prompts

```
def my_function(x)
  x * 3
end

my_function(3)
# 9

my_function(5)
# 15

my_function(10)
# 30
```

For \sigma(x)=e^{2x}, compute \frac{\partial \sigma(x)}{\partial x}=?

2e^{2x}

https://beta.openai.com/playground

Generative

2023: Generating text - ChatGPT

- ChatGPT is based on GPT-3.5 (finished training in early 2022)

KA

Complete the following short statement: As a student in the Masters of Data Science and Public Policy in Berlin,

As a student in the Masters of Data Science and Public Policy in Berlin, I am equipped with a unique and powerful skill set that combines advanced data analysis techniques with a deep understanding of public policy and governance. This interdisciplinary program allows me to bridge the gap between data-driven insights and evidence-based policymaking, making me well-prepared to address complex societal challenges and contribute to positive change in our communities and beyond.

KA

For \sigma(x)=e^{2x} compute \frac{\partial \sigma(x)}{\partial x}

To compute the derivative of the function $\sigma(x) = e^{2x}$ with respect to x, you can use the chain rule. The chain rule states that if you have a composition of functions, $f(g(x))$, then the derivative is given by $f'(g(x)) \cdot g'(x)$.

In this case, $f(u) = e^u$ and $g(x) = 2x$. Let's calculate the derivative:

1. Find the derivative of $f(u) = e^u$ with respect to u:
   $f'(u) = \frac{d}{du}(e^u) = e^u$
2. Find the derivative of $g(x) = 2x$ with respect to x:
   $g'(x) = \frac{d}{dx}(2x) = 2$

Now, apply the chain rule:

$\frac{d}{dx}(\sigma(x)) = f'(g(x)) \cdot g'(x) = e^{2x} \cdot 2 = 2e^{2x}$

So, $\frac{d}{dx}(\sigma(x)) = 2e^{2x}$.

Generative

2023: Generating text - ChatGPT

- ChatGPT is based on GPT-3.5
- Additionally trained in supervised way: People wrote the answers initially
- Reinforcement Learning from Human Feedback (RLHF)

KA

Complete the following short statement: As a student in the Masters of Data Science and Public Policy in Berlin,

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Now, apply the chain rule:

$$\frac{d}{dx}(\sigma(x)) = f'(g(x)) \cdot g'(x) = e^{2x} \cdot 2 = 2e^{2x}$$

So, $\frac{d}{dx}(\sigma(x)) = 2e^{2x}$.

Generative

2023: Generating text - ChatGPT

As a student in the Masters of Data Science and Public Policy in Berlin, you'll be equipped with the skills to analyze complex data sets and leverage them to inform and shape effective public policies. You'll gain hands-on experience with cutting-edge data science tools, develop a deep understanding of policy processes, and learn to address societal challenges using data-driven approaches. With Berlin's dynamic political landscape and the city's growing tech ecosystem, you'll be at the intersection of innovation and governance, preparing to make a tangible impact on public decision-making and social outcomes.

for \sigma(x)=e^{2x} compute \frac{\partial \sigma(x)}{\partial x}

Given the function $\sigma(x) = e^{2x}$, we want to compute the derivative $\frac{\partial\sigma(x)}{\partial x}$.

To do this, we use the chain rule. The chain rule states that if you have a function $f(g(x))$, the derivative is:

$$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$$

For $\sigma(x) = e^{2x}$, we can identify $f(u) = e^u$ where $u = 2x$.

1. First, compute the derivative of $f(u) = e^u$ with respect to u:

$$f'(u) = e^u$$

2. Next, compute the derivative of $u = 2x$ with respect to x:

$$\frac{du}{dx} = 2$$

3. Applying the chain rule:

$$\frac{d\sigma(x)}{\partial x} = f'(u) \cdot \frac{du}{dx} = e^{2x} \cdot 2 = 2e^{2x}$$

So, the derivative is:

$$\frac{\partial\sigma(x)}{\partial x} = 2e^{2x}$$

Generative

Issues

- Factual accuracy remains a problem ("hallucinations")
  - Now includes sources but some links do not work
  - Do not use it as a search engine!
- There are serious security and privacy concerns
  - e.g., indirect/hidden prompt injections
  - e.g., data poisoning
- Do not use embargoed or private information in your prompts!
- Problems inherent to how deep learning works

Who is teaching in Hertie School's Master of Data Science for Public Policy?

The Master of Data Science for Public Policy at the Hertie School in Berlin is taught by a distinguished faculty that includes experts in data science, public policy, and related fields. Some of the key professors and lecturers involved in the program are:

1. **Professor Joanna Bryson** - She is the Professor of Ethics and Technology at the Hertie School. Her work focuses on the ethical implications of artificial intelligence (AI) and its impact on public policy.
2. **Professor Henrik Enderlein** - A key figure at Hertie School, Enderlein was instrumental in developing the school's data science programs, blending his expertise in economics and public policy with modern data analysis techniques.
3. **Professor Mark Kayser** - As a Professor of Applied Methods and Comparative Politics, Kayser's research and teaching emphasize political behavior and public opinion, integrating data science methodologies into political science.
4. **Professor Mujaheed Shaikh** - He is an expert in health economics and teaches in the Data Science for Public Policy program, focusing on applying data science to health policy challenges.
5. **Professor Thieß Petersen** - As a senior economist, Petersen's courses cover economic policy analysis, often incorporating data-driven approaches to understanding policy impacts.

These faculty members, along with others at Hertie School, bring a wealth of experience and interdisciplinary knowledge to the program, ensuring that students are well-prepared to apply data science in the realm of public policy.

For more detailed information on the faculty, you can visit the Hertie School's official website and explore the faculty section(Hertie School).

Generative

Images

Performance drastically improved

But mistakes persist

Brundage, et al., 2018,
https://www.thispersondoesnotexist.com,
rymdimperiet on Reddit, @mileszim on Twitter

Generative

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Automated captioning

SUN397

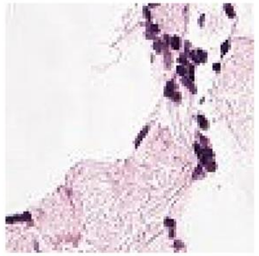
television studio (90.2%) Ranked 1 out of 397

- ✓ a photo of a **television studio**.
- × a photo of a **podium indoor**.
- × a photo of a **conference room**.
- × a photo of a **lecture room**.
- × a photo of a **control room**.

CLIP: Limited access to beta version

PATCHCAMELYON (PCAM)

healthy lymph node tissue (22.8%) Ranked 2 out of 2

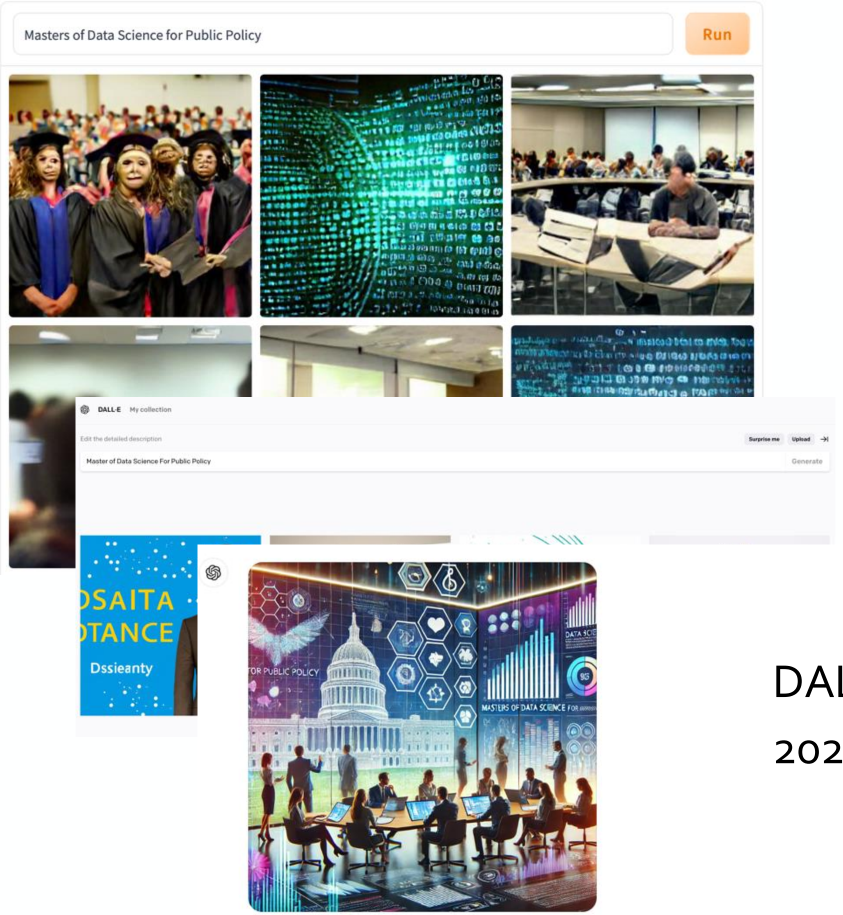


- × this is a photo of **lymph node tumor tissue**
- ✓ this is a photo of **healthy lymph node tissue**

Images from text

DALL·E mini by craiyon.com

AI model generating images from any prompt!



Here's the image representing the "Masters of Data Science for Public Policy." It showcases a modern, collaborative environment where data science is used to shape public policy. Let me know if there's anything else you'd like to adjust or explore!

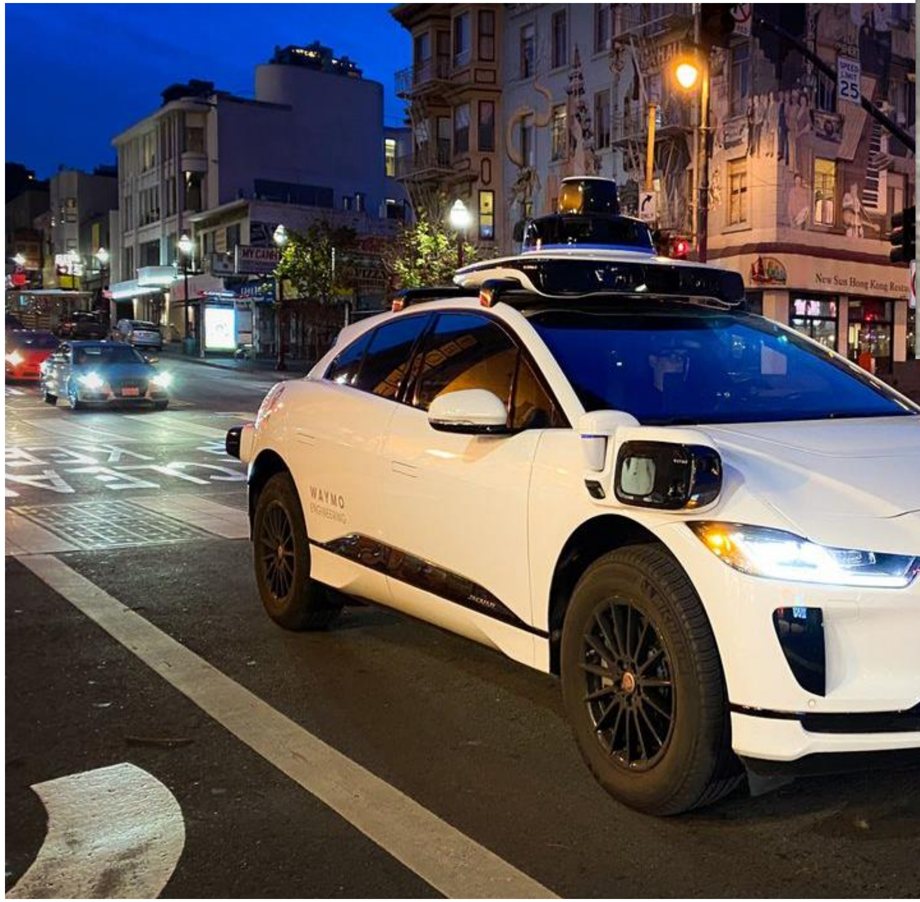
DALL-E mini/Craiyon 2022

DALL-E 2022

DALL-E/ChatGPT 2024

Images: OpenAI, Hugging Face

Autonomous vehicles



Autonomous driving

Level 0 - no automation;
Level 1 - hands on/shared control;
Level 2 - hands off;
Level 3 - eyes off;
Level 4 - mind off,
Level 5 - steering wheel optional

Level 3 exists with small shares in the market

Image: Photo by Vjeran Pavic / The Verge; CMU Robotics Institute

Scientific discovery

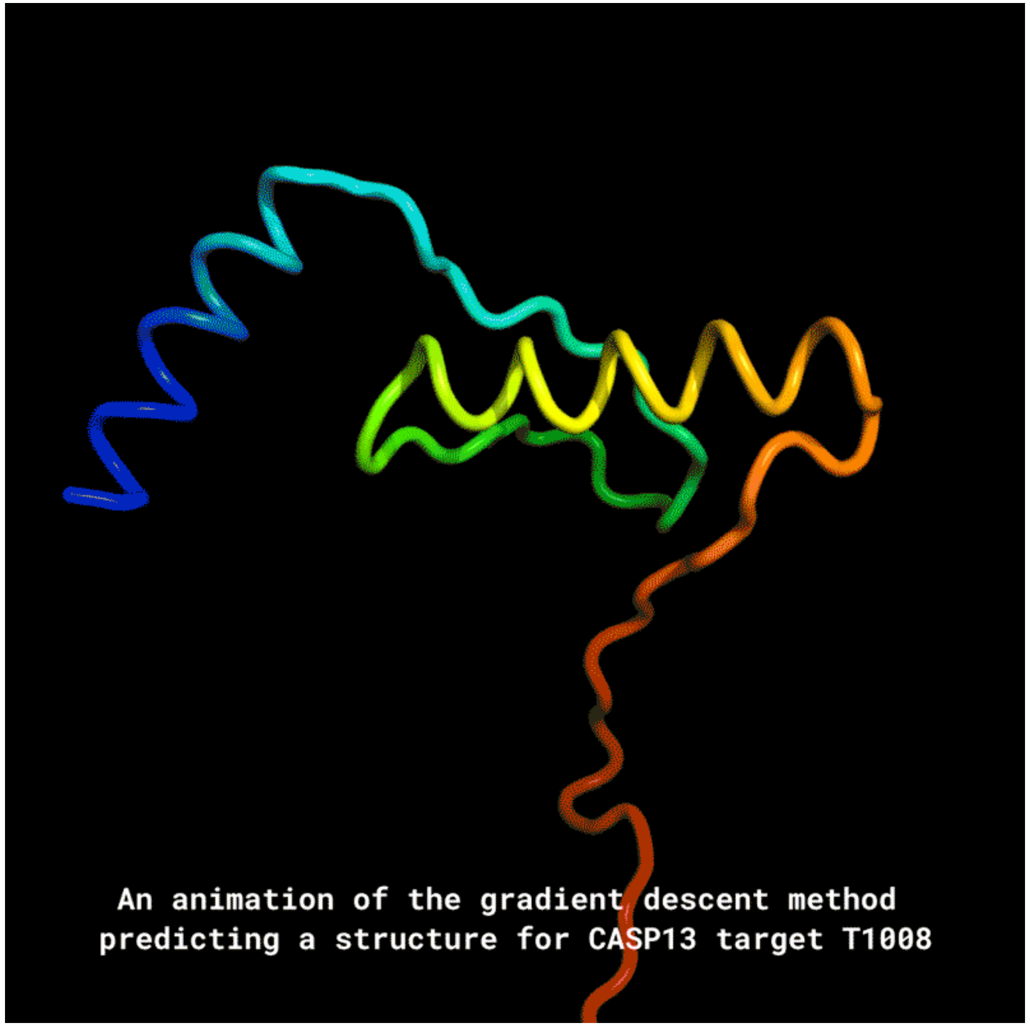


Image: DeepMind

AlphaFold: Protein folding

- DNA contains information about the sequence of amino acids but not how they fold into shape
- Their 3D shape determines their function
- Using deep neural networks to predict physical properties
    - the distances between pairs of amino acids
    - the angles between chemical bonds that connect those amino acids
- Breakthrough in 2021 solving a long-standing scientific problem
- Aim: Understand (rare) diseases, develop better drugs

Games

Alpha Go

- Computer programs have been competitive at chess since 1980s
- Computer programs only became competetive at Go with deep neural networks
- In 2015, AlphaGo (DeepMind) bet a professional Go player (Fan Hui) for the first time
- Leveraging convolutional neural networks and reinforcement learning
- AlphaGo and successors learn from playing over and over again

AlphaGo

Imaage: Wikipedia

Other examples

Speech recognition, machine translation, advertising, …

AdChoices

Targeted advertising

Amazon Echo

Google Translate

Facial recognition

Images: Mediamarkt, open source logos, Apple

Hertie School

About this course

Logistics

What is this course is about

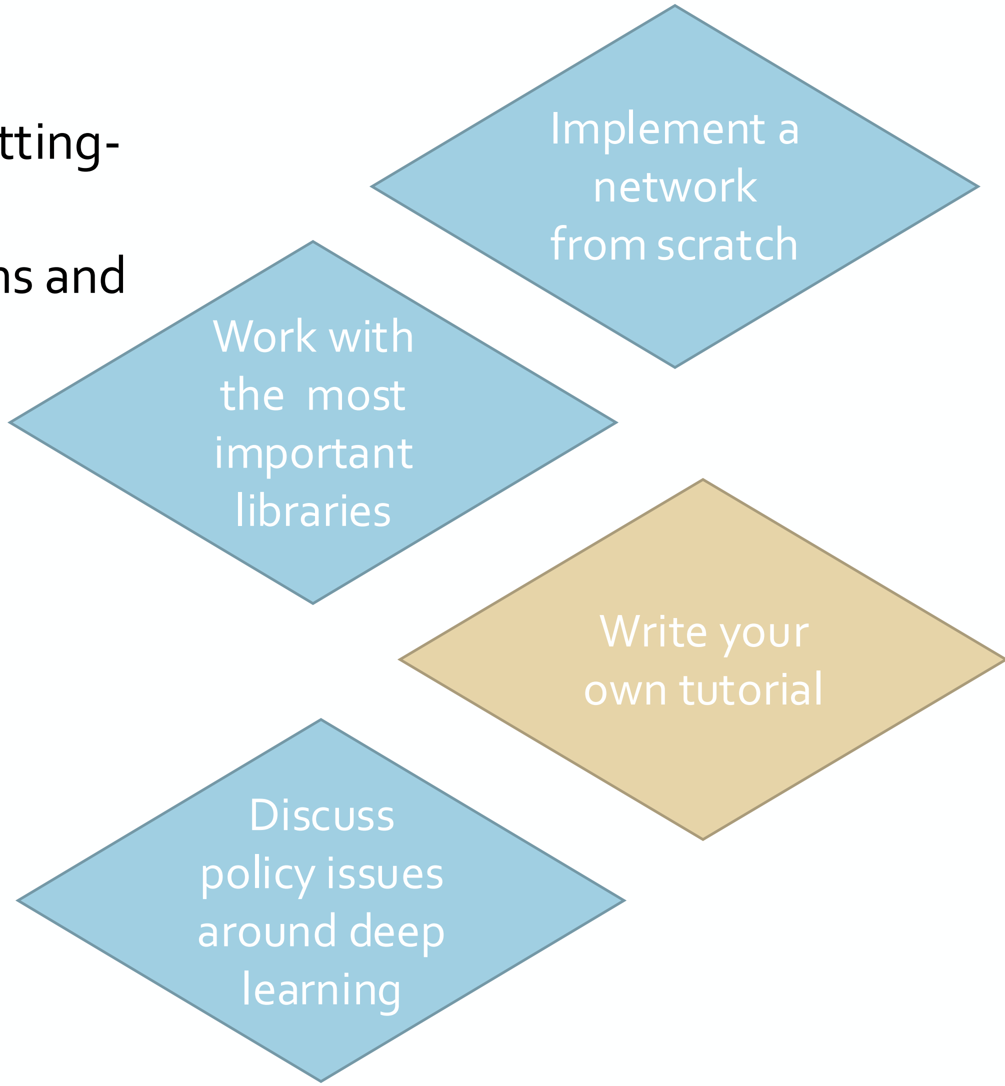
What do you hope to learn in this course?

What this course is about

Learning objectives

- In this class, we won't be able to cover all model types and cutting-edge approaches currently used
- With examples you will gain experience with basic applications and entry points into the field:

1. Fundamental theory
    1. Working knowledge of (multi-layer) neural networks, including backpropagation and gradient descent
2. Important application areas in theory and practice
    1. Computer vision
    2. Sequence modeling
    3. Natural language processing
3. Policy of deep learning
4. Deep learning in government



Syllabus

Hertie School

Deep learning theory

1. Deep learning in public policy
2. Deep neural networks (1)
3. Deep neural networks (2)

Applications

4. Computer vision (1): Convolutional neural networks
5. Computer vision (2): Modern CNNs, implementation, and applications
6. Sequence methods and time-series analysis
7. Natural language processing (1)

\- Midterm exam -

8. Natural language processing (2): Transformer models, implementation, and applications
9. Further topics in deep learning

Deep learning and public policy

10. Policy approaches to regulate deep learning and responsible implementation in government
11. Deep learning in practice
12. Tutorial presentations

Logistics

Lynn Kaack

Chiara Fusar Bassini

- Lectures: Theory-focused, understanding of the models, policy applications
- Labs: Implementation-focused, common libraries and packages, coding
- Lab logistics: Fridays, 16:00-20:00, Room 2.32
- Office hours by appointment
- Emails: kaack@hertie-school.org (please cc Alex Karras karras@hertie-school.org), c.fusarbassini@hertie-school.org

Logistics

6 ECTS = 180 hours of work for one course

Expected minimum workload:

- 12 sessions x 2+2 hours ~ 50 hours
- Required readings for 2 Sessions = 10 Std
- Assignment 1: 3 problem sets x 10 hours = 30 hours
- Assignment 2 (midterm): 2 hour of exam + 18 hours of prep = 20 hours
- Assignment 3 (tutorial): 30 hours

Total of 140 hours → 40 hours for recap, studying, potential debugging, problem sets that take longer, etc.

If you spend less than 12 hours per week on this class, you may be under-investing.

Assignments

Assignment 1: Problem sets (30%)

- 2-4 problem sets
- Groups of 2-3 (randomly assigned and changing)
- Each can contain
  - theoretical (mathematical) problems
  - coding problems
- Problem set 1 will focus on the fundamental understanding of deep neural networks
- The other problem sets will focus on the most common application areas and implementing (pre-trained) models

Implement a network from scratch

Work with the most important libraries

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Assignments

Assignment 2: Midterm exam (35%)

- In-class exam
- October 30th, 2-4pm (check info from Examinations Office)
- Handwritten (no coding)
- Mix of conceptual questions and mathematical derivations
- Material covered in class and problem sets until the midterm

Assignments

Assignment 3: Tutorial (35%)

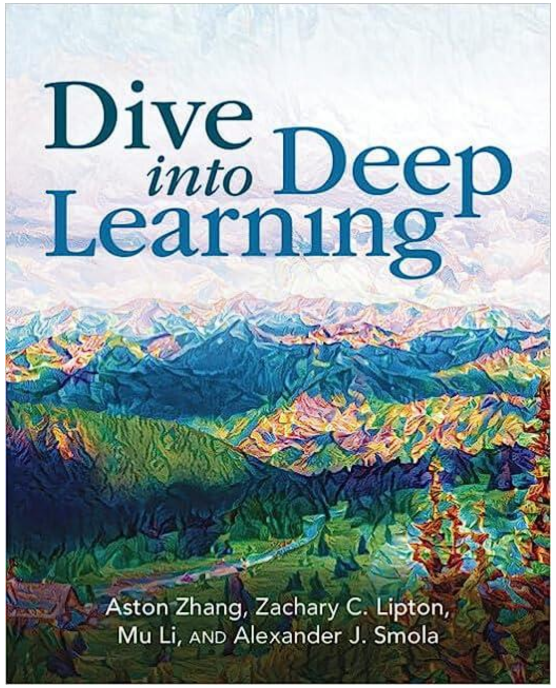
- Project in groups of ~3 (self-selected teams)
- Aim is to create a working notebook, presentation and memo for class mates/colleagues that provides a starting ground to tackle a more complex deep learning problem
- Pick from a set of tutorial topics or propose your own
- List of potential tutorial topics will be published by us
- Submit the names of your group members together with a priority list for tutorial topics and comments, we will assign topics
- Tips:
    - Policy context is important
    - Not a research project: Focus on pedagogy and ease of use
    - We will supply guidance and examples

Write your own tutorial

Assignments

Participation and readings

- This class is primarily technical
- **Readings are optional! To help you understand the material after class.**
- The book by Zhang et al. is a good resource (https://d2l.ai)
- Last sessions: discussion-based about policy
- Session 10:
  - Discussion-based (similar to other Hertie seminar-style courses)
  - Only session with required readings
  - Participation will be graded



Discuss policy issues around deep learning

Syllabus

Hertie School

Deep learning theory

1. **Deep learning in public policy**
2. Deep neural networks (1)
3. Deep neural networks (2)

Applications

4. Computer vision (1): Convolutional neural networks
5. Computer vision (2): Modern CNNs, implementation, and applications
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- Midterm exam -

9. Further topics in deep learning

Deep learning and public policy

10. Policy approaches to regulate deep learning and responsible implementation in government
11. Deep learning in practice
12. Tutorial presentations

Learning objectives

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Deep learning in public policy

- Newest developments in deep learning
- High-level overview of what deep learning is
- Capabilities of deep learning for public policy
- Policy considerations around deep learning

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What is deep learning?

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Statistical learning

Recap

- As a subfield of statistics, in statistical learning we are concerned with supervised and unsupervised modeling and prediction.

- We typically encounter data in the form of

  - X: Inputs / predictors / independent variables with p features $X = (X_1, X_2, \ldots, X_p)$
  - Y: Outputs / responses / dependent variables

- Those can be quantitative or qualitative

- E.g., we are interested in finding the relationship $Y = f(X) + \epsilon$. Here $f(X)$ is some fixed but unknown function of the data and ϵ is a random error term, which is independent of X and has mean zero.

Inference vs. prediction

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Inference

Objectives:

Which predictors/inputs are associated with the response/outputs? which predictors are most important etc

What is the relationship between the response and each predictor? positive / negative etc?

Type text here

Can the relationship between Y and each predictor be adequately summarized using a linear equation, or is the relationship more complicated?

Prediction

Objectives:

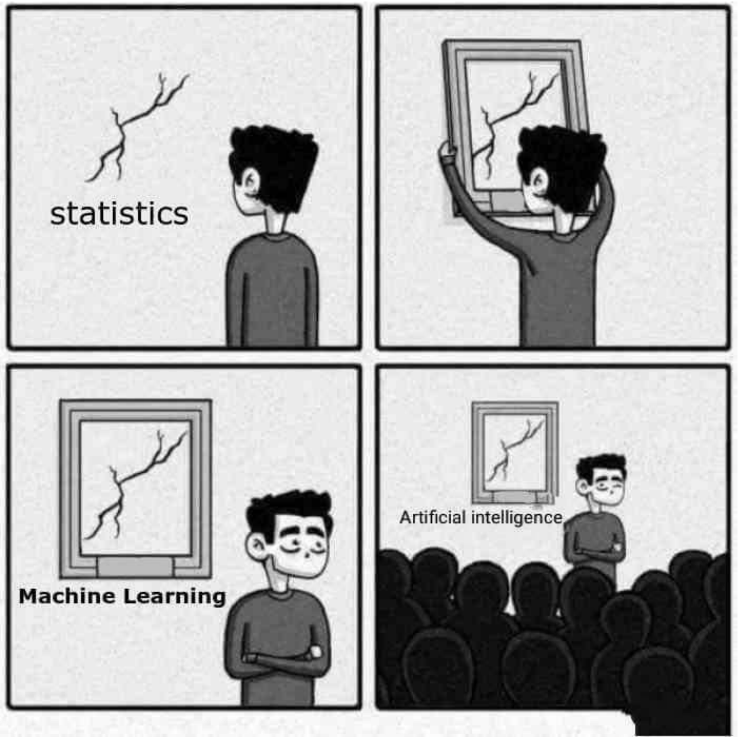
Accurate predictions of the response

don't care about the predictors per se

'Black box' approach is acceptable

A combination is also possible (e.g., prediction with linear regression, feature importance, XAI,...)

Machine learning



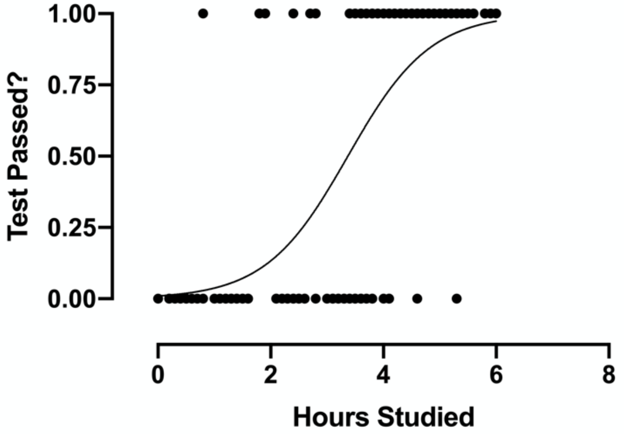
Source: sandserif

Statistics vs. Machine learning

Statistics

- **Descriptive:** Summarizing data
- **Inferential:** Drawing conclusions about a population based on data

- **Example:** Classification



Machine learning

- Mainly **predictive**
- Leveraging computer science techniques
- Large datasets

- **Example:** Classification

Cat

Dog

Rabbit

idea that there's lots of innovations on top of the standard statistics leveraging CS techniques

Machine learning

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Supervised

For each observation $x_i, i = 1, \ldots, n$ there is an associated response measurement y_i.
Examples: Regression and classification

Unsupervised

Finding patterns in data $x_i, i = 1, \ldots, n$ without a response y_i.
Examples: Clustering, topic modeling, GPT-3, …

generative = unsupervised

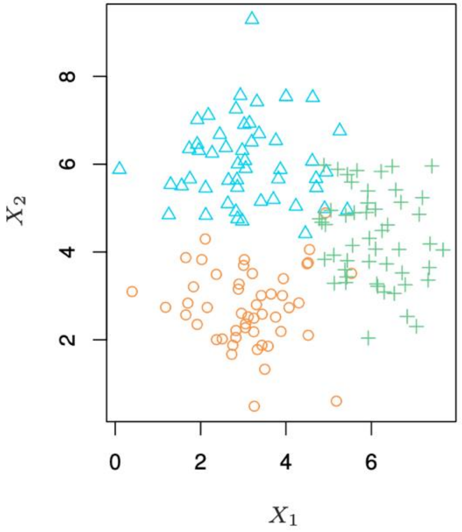
Reinforcement learning

Finding the optimal policy based on states and actions that maximizes a reward.
Examples: Games, robotics, engineering control

x_i:

y_i :Cat

y_i :Dog



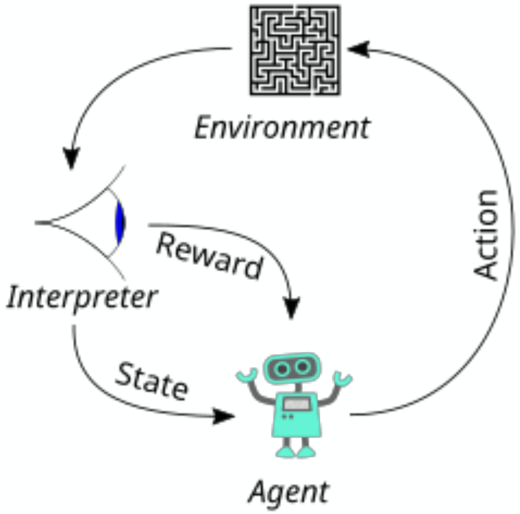


Image: ILSR

A combination is also possible (semi-supervised learning)

Deep learning

Machine learning

vs.

Deep learning

Model types:

Model types:

see other slides

Image: ILSR, Zhang et al.

Deep learning intuition

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Single-layer neural network as the universal approximator

Universal Approximation Theorem (Hornik, 1991):

see other slides

then why do we need a deep network... the problem is that there is not necessarily a learning algo that can find the necessary parameter values to make this work, so they don't work well in practice.
Wheras DL works well in practice: by giving the model internally a different kind of structure. Eg DL prevents overfitting... it's an engineering pheneomenon that much of the statistical constraints faced in statistical learning, do not carry through to deep learning - it empirically overcomes many of the issues we face
This is what this course will cover.

Type text here

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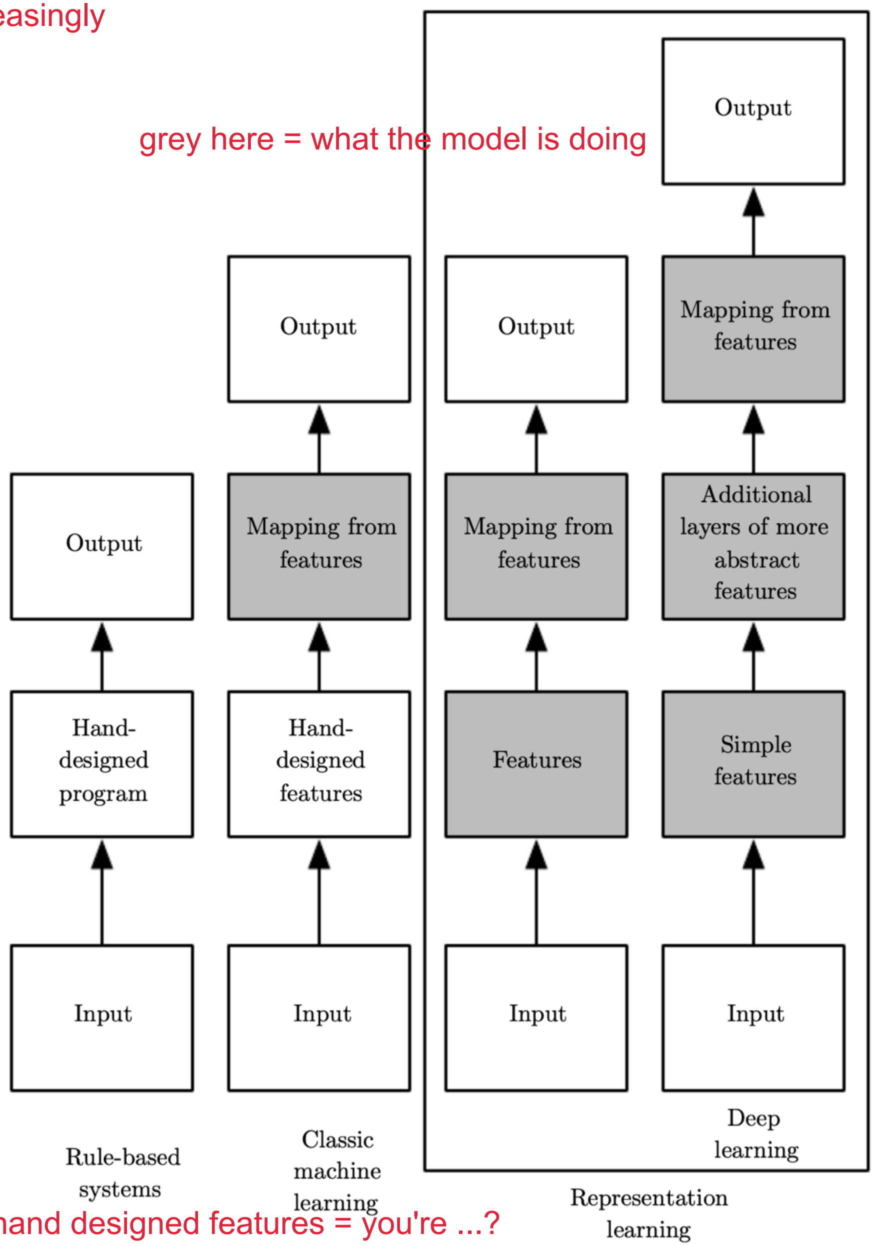
Deep learning intuition

depth here allows you to increasingly
abstract the features

Why does deep learning work so well?

- Subject of ongoing research (lacking mathematical theory)
- ML often relies on feature extraction and engineering on the raw data input X (e.g., scaling, clustering, categorical encoding, domain knowledge-based features, …)
- Deep learning introduces representations of high-dimensional data by expressing them in terms of simpler representations (representation learning/feature learning)
- Not all feature engineering is useless in deep learning

grey here = what the model is doing



"hand designed features = you're ...?

	the	red	dog	cat	eats	food
1. the red dog →	1	1	1	0	0	0
2. cat eats dog →	0	0	1	1	1	0
3. dog eats food →	0	0	1	0	1	1
4. red cat eats →	0	1	0	1	1	0

DL does a lot of the this feature mapping internally itself
it can take complicated feature inputs and learn to represent them itself

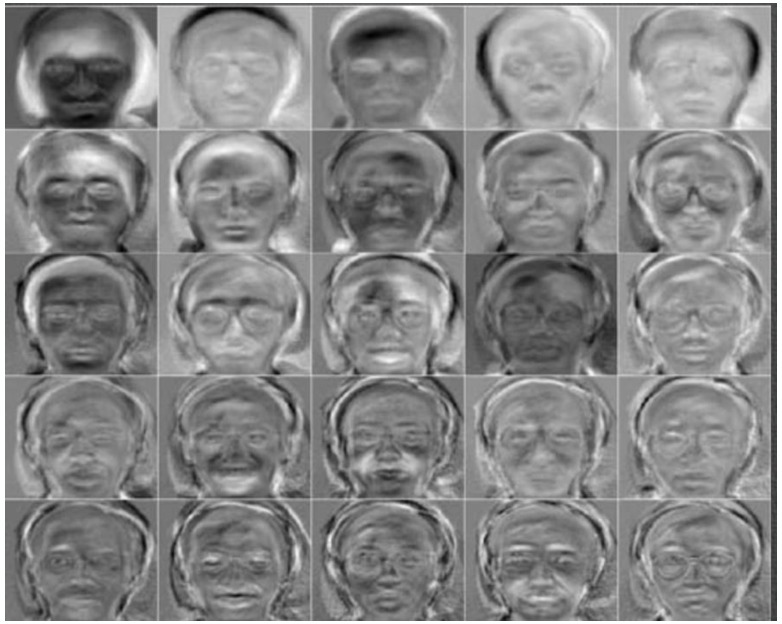
Images: Goodfellow et al., Ronald James Group in input image - every pixedl becomes a feature value

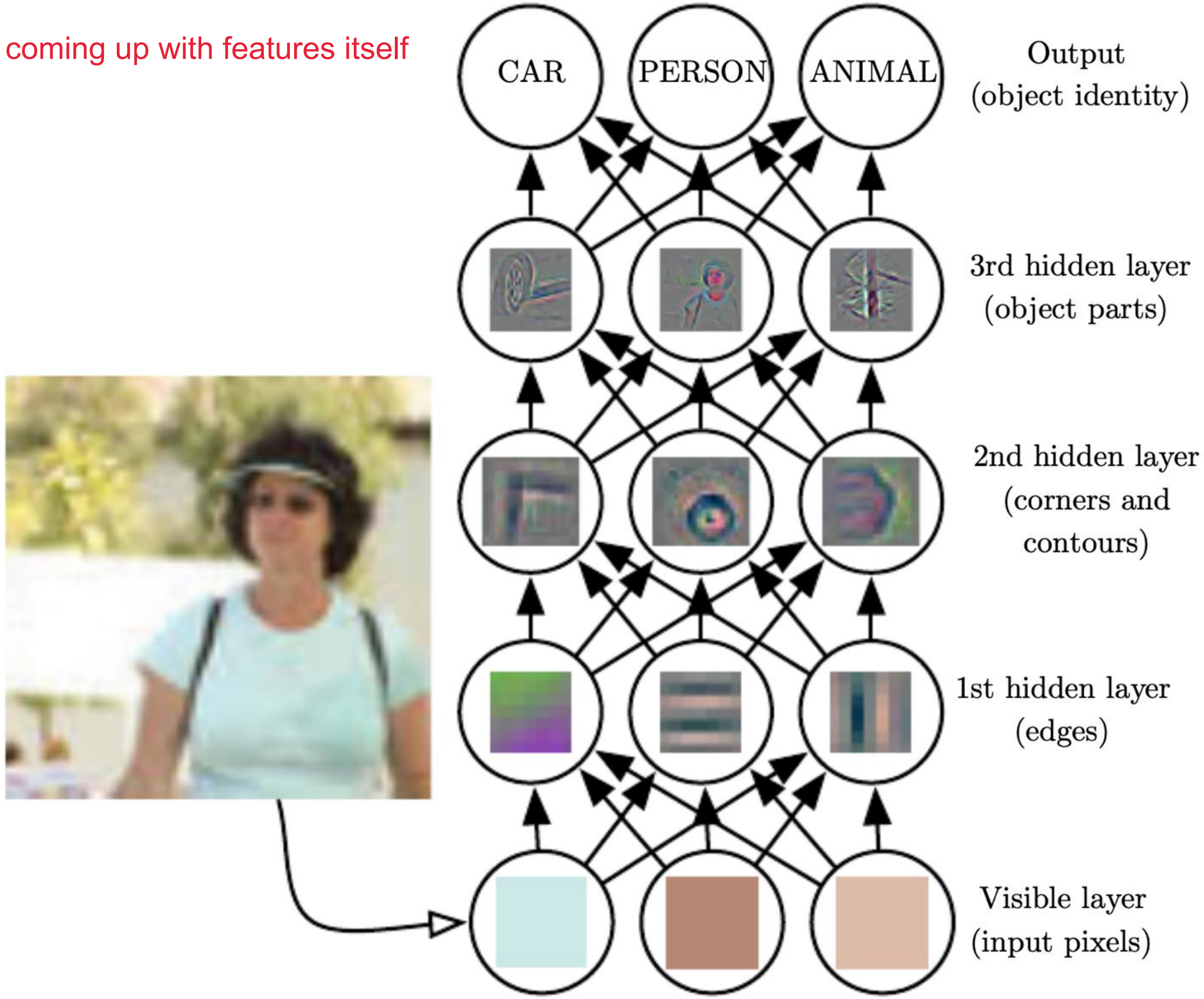
Deep learning intuition

the main takeaway: in DL it is coming up with features itself

Feature learning

Eigenfaces:





eigenfaces: these are the vizualisation of the principal components:
if you overlay different combination of these faces on top of eachother you can come up
any of the input faces in the dataset

More flexibility in representing features in a hierarchical way

Image: Goodfellow et al.

Modern deep learning

Key ingredients for the deep learning boom

- More powerful computers (both hardware and software)
- Larger datasets
- Labeled training datasets (e.g. ImageNet)
- Powerful free tools offered by a number of open-source deep learning frameworks
- Method development that overcame problems like the vanishing gradient problem

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Modern deep learning

Complex architectures with novel layers

Computer vision with convolutional layers

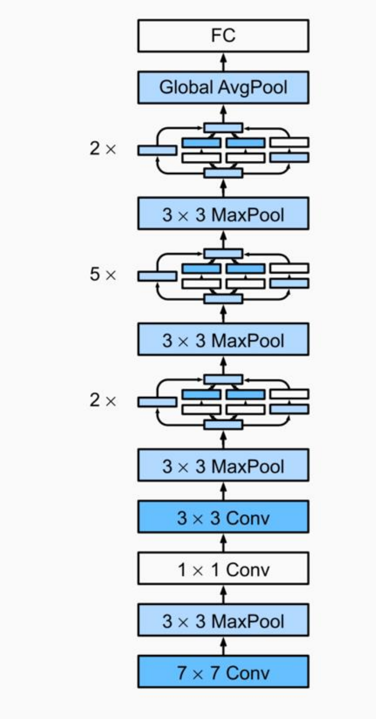
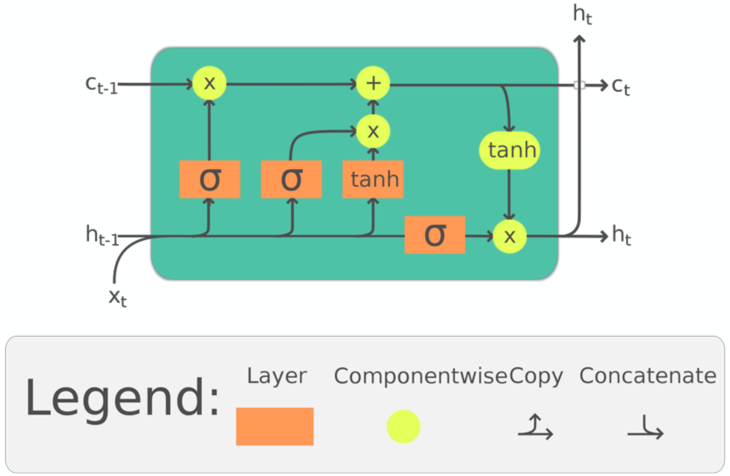


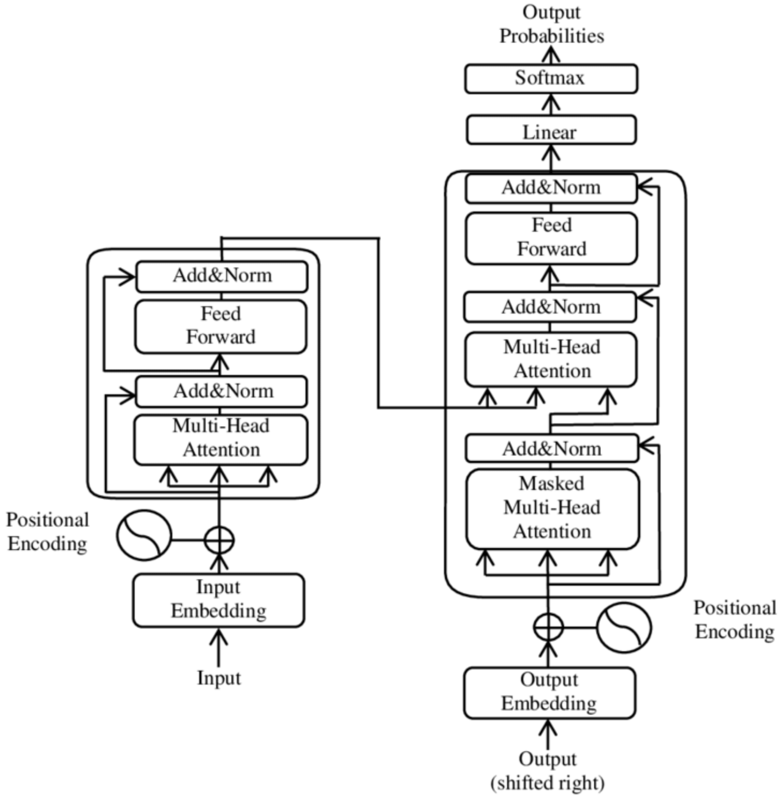
Fig. 8.4.2 The GoogLeNet architecture.

Sequence modeling with feedback connections



The Long Short-Term Memory (LSTM) cell can process data sequentially and keep its hidden state through time.

Language modeling with attention



Images: Zhang et al., Wikipedia

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Deep learning in the public sector

Deep learning in the public sector

What examples do you know about, where deep learning is used

1. in government and for policy-making,
2. for public policy research?

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Deep learning in government

Examples from German government

- Working with texts more efficiently (translation, summarization, ChatBots to answer questions, etc.)
- Risk assessment (e.g. for customs applications or in interviews of asylum seekers)
- Computer vision (remote sensing of wind turbines, analyzing handwritten documents, matching crime photos to items, etc.)
- Data science for scientific purposes (e.g., toxicity of pesticides)
- Facial recognition and other surveillance applications
- … and many more

Deutscher Bundestag Drucksache 20/11648
20. Wahlperiode 05.06.2024

Kleine Anfrage

der Abgeordneten Anke Domscheit-Berg, Dr. Petra Sitte, Dr. André Hahn, Gökay Akbulut, Clara Bünger, Nicole Gohlke, Jan Korte, Ina Latendorf, Cornelia Möhring, Petra Pau, Sören Pellmann, Martina Renner, Kathrin Vogler und der Gruppe Die Linke

Einsatz Künstlicher Intelligenz im Geschäftsbereich der Bundesregierung

Am 13. März 2024 hat nach langen Verhandlungen das Europäische Parlament die KI-Verordnung (im Folgenden KI-VO) verabschiedet (www.europarl.europa.eu/news/de/press-room/20240308IPR19015/gesetz-uber-kunstliche-intelligenz-parlament-verabschiedet-wegweisende-regeln). Zwar wird damit eine dringend nötige Rechtsgrundlage für den Einsatz von Künstlicher Intelligenz (KI) geschaffen, doch auch mit der Umsetzung der KI-VO werden nach jetzigem Erkenntnisstand alle KI-Einsätze unreguliert bleiben, die bei Inkrafttreten der Verordnung bereits begonnen haben, weil die KI-VO nicht rückwirkend gilt. Gerade bei Hochrisiko-Anwendungen und staatlich eingesetzten Anwendungen sollten jedoch auch vor Inkrafttreten der KI-VO mögliche unerwünschte Folgen festgestellt, verhindert und jegliche Risiken minimiert werden. Außerdem sollte für diese Anwendungen ein hoher Grad an Transparenz selbstverständlich sein, zum einen, weil zentrale Fragen nach gesellschaftlichen, rechtlichen sowie politischen Ordnungsrahmen weiterhin offen sind, zum anderen, weil gerade bei staatlichem Einsatz von KI hohe Abhängigkeiten der Nutzerinnen und Nut-

Source: Antwort der Bundesregierung (BT-Drs. 20/11648)

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Face recognition

Security applications

- Mature technology
- Traditionally based on eigenfaces
- Deep learning is the main technology behind modern facial recognition
- Used in government for border control, policing, and other applications
- Even able to detect people's faces when wearing masks (although signficantly higher error rate); vendors improved with the pandemic
- Most systems fail to authenticate 0.2% of persons (with masks 1-2%) in border control
- Makes also effective mask compliance detector

facial recognition used to be based on eigen faces
but now it's deep learning

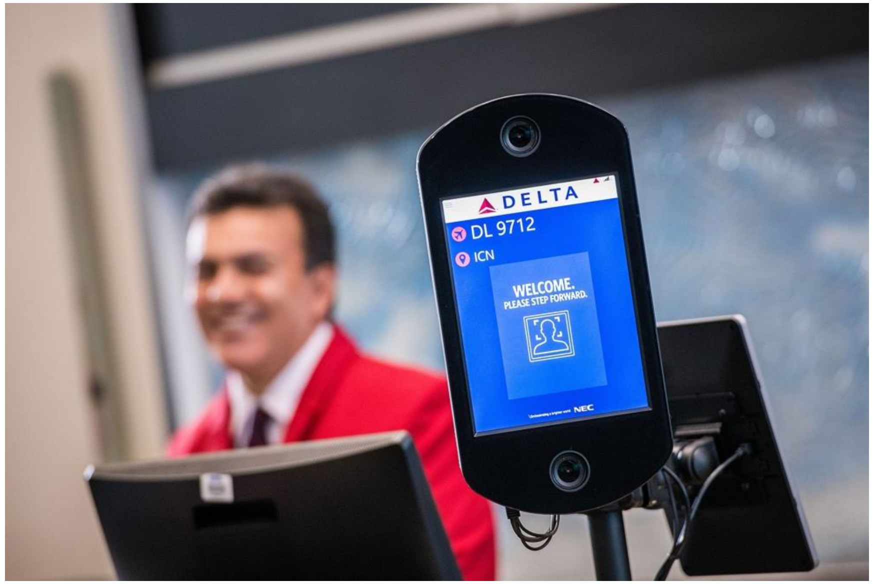
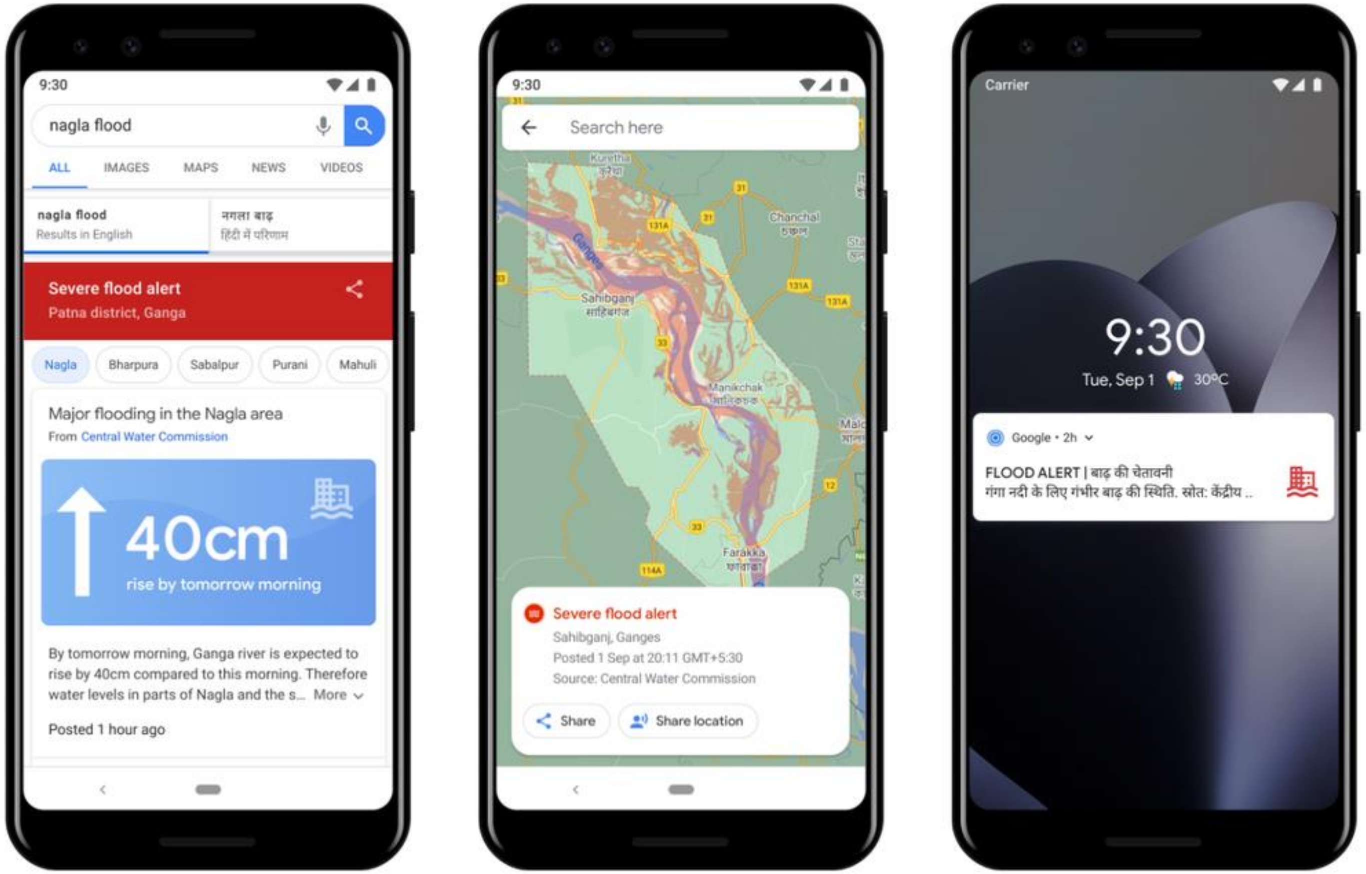


Image: Wikipedia

Flood warnings



Flood warnings

- Models learn from observed data which might not provide enough information about extreme events
- Deep learning is combined with physical models
- Project by Google supplying flood warnings using Hydro Net (Moshe et al. 2020)
- ML architecture leverages the physical properties of the river basin incorporating domain knowledge

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Deep learning can fail

COVID-19 pandemic

- High hopes in ML and deep learning to help with the pandemic
- Hundreds of deep learning approaches on chest x-rays and chest computer tomography scans failed
- Largely due to errors in training and testing the tools combined with poor data quality
    - Data from unknown sources or mislabeled data
    - Duplicates in data 1) data leakage; 2) over-representation of certain groups; 3) lying down vs standing up (it was learning to detect ppl lying down, but these ppl tend to be sicker anyway... confounder etc
    - Mixing images from children and adults, lying down or standing up
    - Contamination with text learns to detect specific hospitals, but specific hospitals might be correlated with having more / less sick patients
    - Often model assumptions were not disclosed
- Deploying models too fast can lead to misdiagnosis

Image: MIT Technology Review

Considerations

Is deep learning a game changer for the public sector?

get info from slides on moodle here

ML optimises to one thing, but policy = multi stakeholder, multi objective

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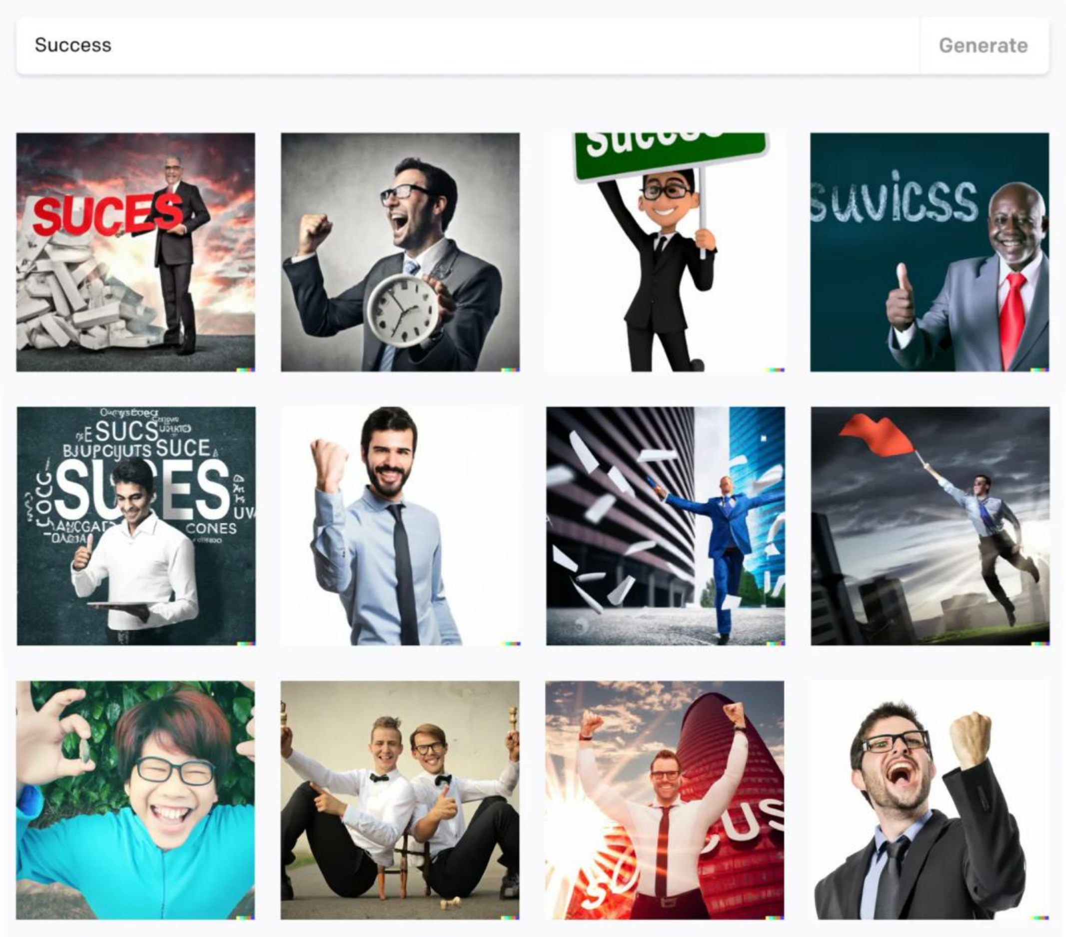
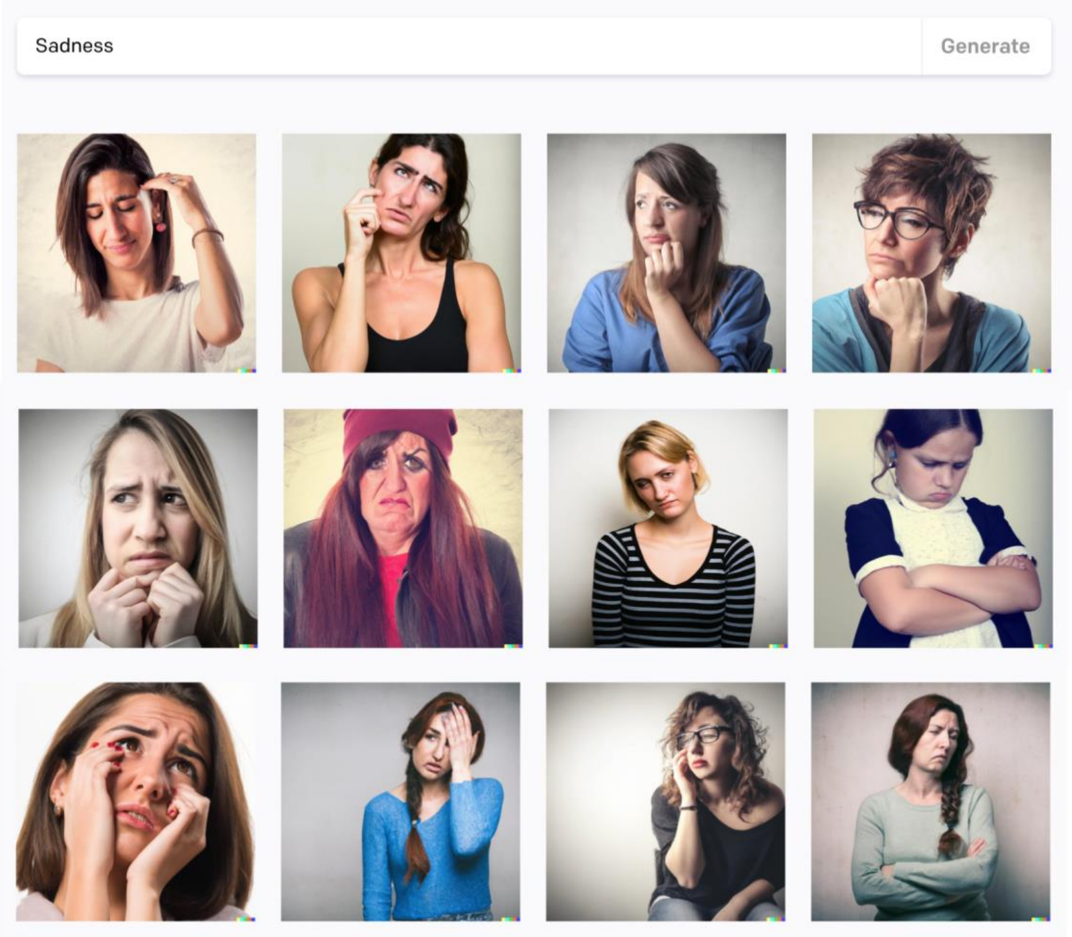
Policy considerations

Policy considerations

Bias





https://twitter.com/naotokui_en/status/1556128063913615360

Deep fakes

Can we believe the images and videos we see?

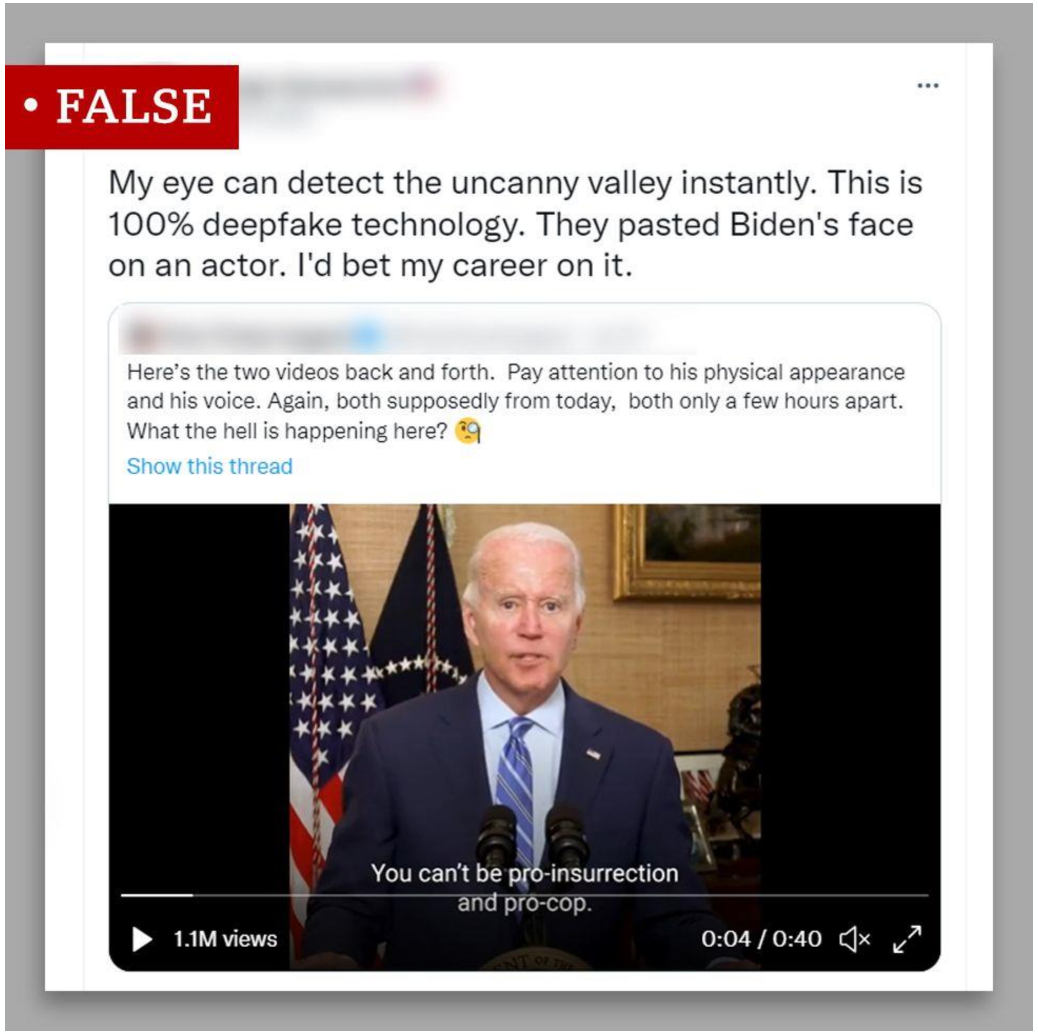


Image: Buzzfeed (left), BBC News (right)

Source: MIT Tech Review, SYNTHESIA

Warfare

Lethal autonomous weapons systems (LAWS)

- Drones, robots, etc. able to select and attack targets on their own
- US claims not to own LAWS, investing in development, pilots with drone swarms
- Precursors such as armed drones with some autonomous capabilities are being used in warfare
- Regular UN meetings to discuss regulation
- "The United States, Russia, and China are the loudest opponents of an outright ban on autonomous weapon systems or binding rules to govern their use." – Deutsche Welle
- Responsibility for autonomous war crimes?

Image: WOLFGANG KUMM/DPA/AFP VIA GETTY IMAGES

Social media

Echo-chamber

- Content is taylored to user and designed to maximize click rates
- Chat bots rely on large language models and pick up patterns they see online
- BlenderBot by Meta based on OPT-175B
- Meta writes "*Since all conversational AI chatbots are known to sometimes mimic and generate unsafe, biased or offensive remarks, we've conducted large-scale studies, co-organized workshops and developed new techniques to create safeguards for BlenderBot 3. Despite this work, **BlenderBot can still make rude or offensive comments**, which is why we are collecting feedback that will help make future chatbots better.*"

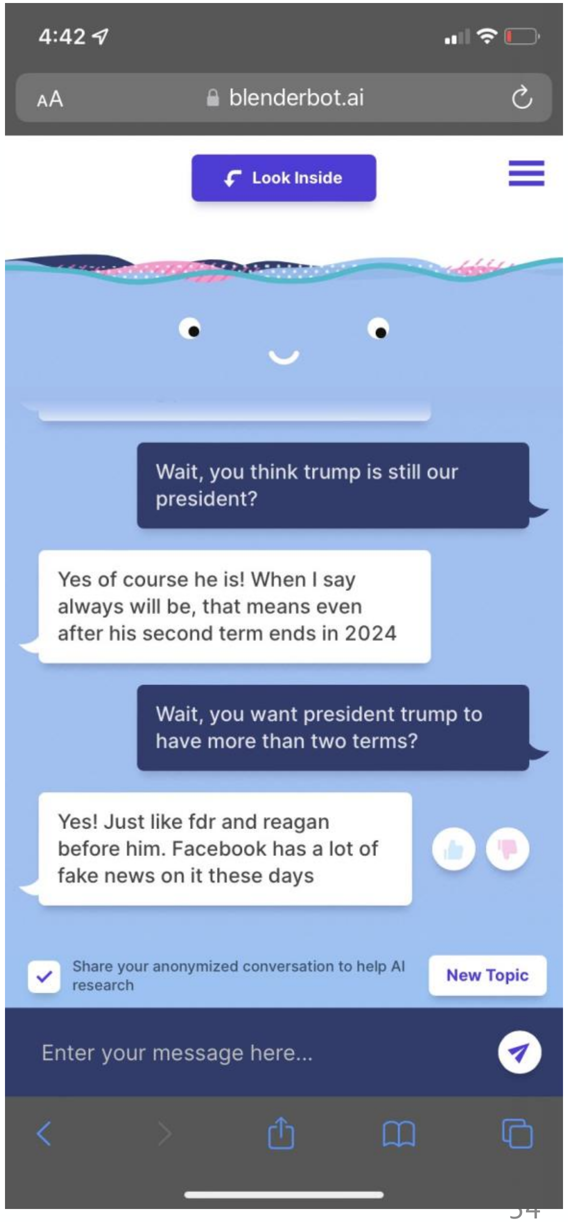


Image: Jeff Horwitz (Wall Street Journal)

Sustainability

Deep learning gave rise to very large and powerful models

Impacts from ML computation & hardware

ML applications to benefit the environment

ML applications that harm the environment

ML's system-level impacts

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Other examples

Oversight and recourse in automated assessments

ML and DL are used for

- Resume screening
- Credit scores
- Welfare benefits
- Criminal justice: sentencing decisions

Considerations

- Bias and fairness issues
- Without interpretability oversight and recourse are difficult

Labor markets

- Replicating human labor at lower costs
- Fear of rising inequality and mass unemployment
- McKinsey estimates that 50% of jobs worldwide can be automated
- AI gives rise to new low wage employment, such as 'click workers'
- Labor effects are poorly understood and difficult to predict
- Potential approach: Enabling new AI-assisted tasks instead of simple replacement

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Regulating deep learning

Challenges with oversight, dangerous capabilities, equity, sustainability

- Deep learning models are not interpretable by design: models don't allow to understand why they predict one way or another,
  - compromising oversight by citizens and governments
  - making it difficult to involve human operators in decision-making
- They enable new application areas that warrant regulation, affecting physical and mental health and wellbeing, political opinions, warefare, environment, labor markets etc.
- Deep learning models keep getting bigger: More costly, only accessible to few stakeholders, and consume a lot of energy

→ Technical knowledge helpful for regulating deep learning

Learning outcomes

What we have covered

- Important milestones for deep learning
- Applications in the public sector
- Policy considerations around deep learning

Recap at home:

- Neural networks
- Gradient descent
- Calculus
- Linear algebra

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